Sinusoidal Distribution

Baidurya Bhattacharyya
MD2403, M. Stat. 2nd Year
under the supervision of
Dr. Ayanendranath Basu,
ISRU, ISI Kolkata.

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Abstract

We introduce the *Sinusoidal distribution*, a flexible 4-parameter location-scale-skewness-kurtosis family of distributions, based on Gilbert's Sine distribution (1893). The mathematical properties of this distribution are studied. Examples of fitting the distribution to multiple statistical objects are considered to demonstrate the distribution's flexibility. The practical utility of the distribution is demonstrated by considering its aptitude at the tasks of fitting, modelling test statistic distributions, modelling linear regression errors and serving as a flexible family of activation functions in neural networks. Future prospects as well as hindrances for each of these avenues is discussed.

Contents

Chapter 1

Introduction

There is significant need for flexible probability distributions in statistical domains. Although most frequently used densities are unimodal and "bell shaped", their shapes can be arbitrarily complex. Several new candidates, such as Chalabi, Scott, and Wuertz 2012, and at the same time, multiple new techniques, some of which are summarized in Anusha, Bhatracharyulu, and Jayasree 2024, have been devised to generate more flexible shapes from existing distributions.

One instinctive choice for a bell-shape is the crest of a sine-curve. Even disregarding its elegant mathematical properties, this crest forms a visually natural candidate as a probability density function. Gilbert 1893 mentions the *Sinusoidal distribution* in its most well recognized form to study the distribution of moon-craters, although in Edwards 2000 we find a brief history of the same density expression in the works of La Grange 1770 and Pearson 1902.

Several attempts have been made to extend the Sine Distribution into a more flexible array of shapes to model a wide variety of data distributions, including SS transformed Sine distribution (Kumar, U. Singh, and S. K. Singh 2015), Sine-G distribution (Mahmood, Chesneau, and Tahir 2019), and Sine generalized family (Oramulu et al. 2024) although such recent modifications propose formulatically complex extensions of Gilbert's proposed elegant density. In our work, we consider a simple 4-parameter extension of Gilbert's baseline Sine Distribution into a flexible, location-scale-skewness-kurtosis family, that is equally elegant in its density expression to Gilbert's Sine Distribution.

We navigate through the work as follows: at first, we introduce the *Sinusoidal distribution* and investigate its shape properties. We will then take a look at some immediate mathematical results that follow from its density expression. Then we take a deeper look at its mathematical properties using the notion of Generalized Hypergeometric Functions. Then, we provide instances of the flexibility of the *Sinusoidal distribution* by highlighting its fitting capabilities. Then we discuss its possible applications in data-modelling, test-statistic modelling, regression-error modelling and in the domain of machine learning as the activation function of a Neural Network.

Chapter 2

Sinusoidal Distribution

2.1 The Sinusoidal Distribution

First we introduce the "Standard" *Sinusoidal distribution* family as follows:

Random variable Z is said to follow a Standard *Sinusoidal distribution* when it is distributed according to the density:

$$\boxed{\psi(z; s, k) \equiv \frac{\varsigma(z; s, k)}{A(s, k)} = \frac{1}{A(s, k)} \sin^k(\pi z^s); \quad z \in [0, 1]} \tag{2.1}$$

where the parameters are $s \geq 0, k \geq 0$, the kernel is notated as $\varsigma(z; s, k) := \sin^k(\pi z^s)$, and $A(s, k) := \int_0^1 \sin^k(\pi t^s) \mathrm{d}t$ is the normalizing constant, i.e. the complete integral of the kernel over $[0, 1]$. The notation for the distribution is $Z \sim \mathrm{Sinu}(s, k)$, and naturally, its CDF is given by:

$$\Psi(z; s, k) = \int_0^z \psi(t; s, k) \mathrm{d}t \equiv \frac{A_z(s, k)}{A(s, k)}; \quad z \in [0, 1] \tag{2.2}$$

where $A_z(s, k)$ denotes the incomplete integral of the kernel.

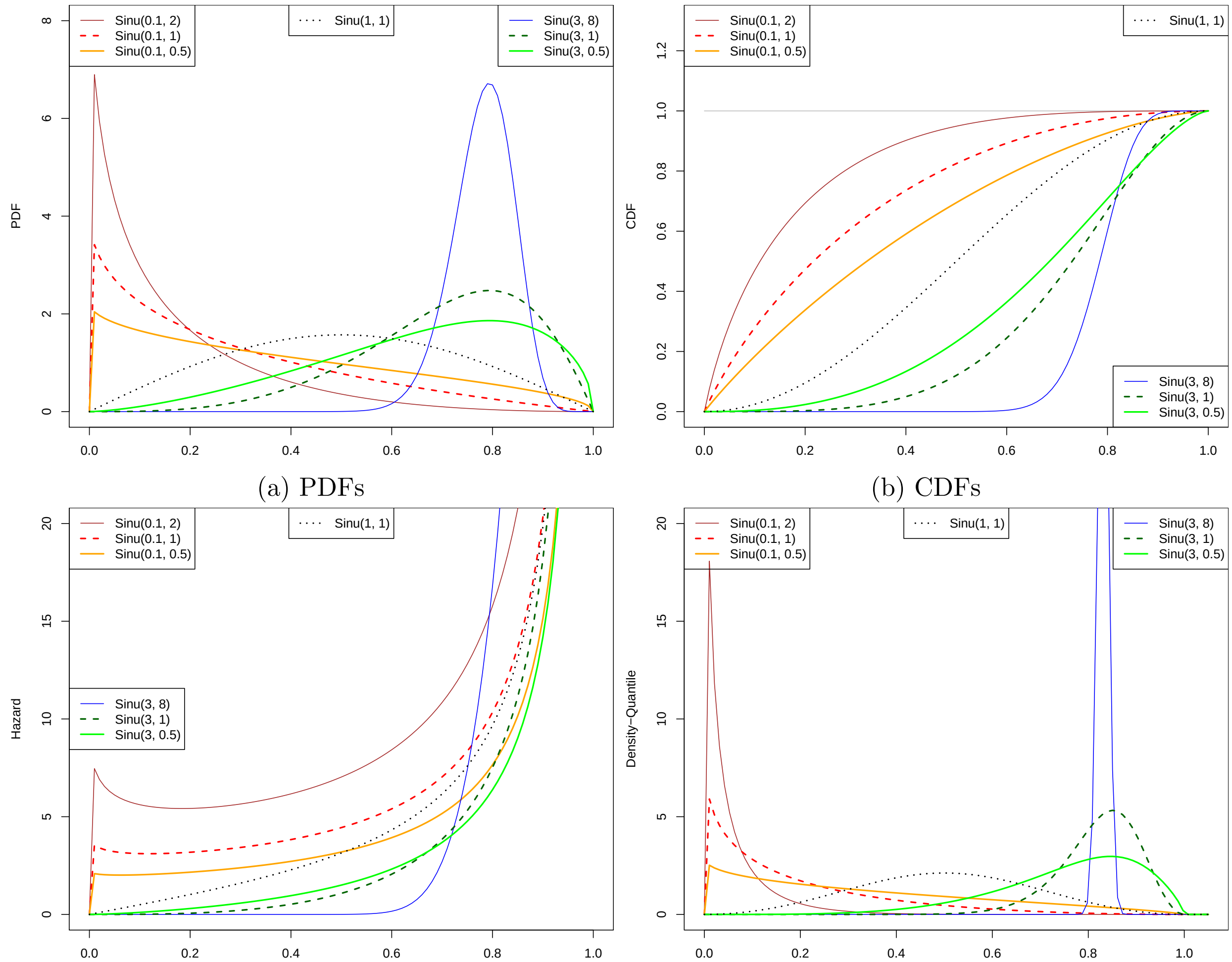
We now introduce 2 extensions. Firstly, the location scale extension is defined as follows: random variable X follows a location-scale *Sinusoidal distribution* family, notated $X \sim \mathrm{Sinu}(a, d, s, k)$ with parameters $a \in \mathbb{R}, d \geq 0, s \geq 0, k \geq 0$, when X has density:

$$\psi(x; a, d, s, k) = \frac{1}{d} \psi\left(\frac{x-a}{d}; s, k\right); \quad x \in [a, a+d] \tag{2.3}$$

Next, we introduce a "flipped extension" (for the reason described in subsection 2.1.1, item 5) to the *Sinusoidal distribution*: random variable X follows a flipped location-scale *Sinusoidal distribution* family, notated $X' \sim \mathrm{Sinu}'(a, d, s, k)$, when $X' :\stackrel{d}{=} a + d - X$ where $X \sim \mathrm{Sinu}(a, d, s, k)$. Conceivably, X' has density

$$\psi'(x; a, d, s, k) = \frac{1}{d} \psi\left(\frac{a+d-x}{d}; s, k\right) \tag{2.4}$$

Figure 2.1 illustrates the PDFs, CDFs, Hazard functions and DQFs (Density-Quantile Func-



(a) PDFs

(b) CDFs

(c) Hazard Functions

(d) DQFs

Figure 2.1: Shape plots of different *Sinusoidal distributions*

tions, as described in Staudte 2017) of some Standard *Sinusoidal distributions*.

2.1.1 Shape Properties

Some shape properties of the *Sinusoidal distribution* are discussed below:

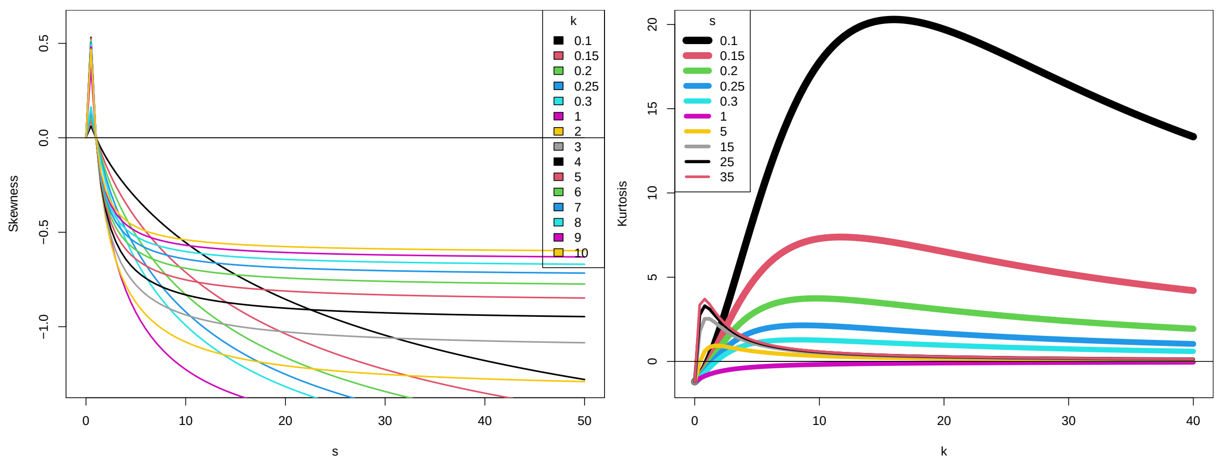
1. Support: $[a, a+d]$. a is the location and d is the scale parameter. Alternative parametrization with $b = a + d$ may be considered.

2. s affects skewness of the *Sinusoidal distribution* as follows:

   (a) $0 < s < 1$: left-leaning (positively/right skewed)

   (b) $s = 1$: symmetric

   (c) $s > 1$: right-leaning (negatively/left skewed)



(a) Skewness vs. s (b) Kurtosis vs. k

Figure 2.2: s, k jointly affect skewness & kurtosis

The alternative parametrization $s' = \ln s$ calibrates the parameter choice $s = 0$ to the symmetric family, and on either side of it keeps oppositely signed skewness families.

3. The choice $a = -\frac{d}{2}$, $s = 1$ gives the special symmetric around 0 $\text{Sinu}(-\frac{d}{2}, d, 1, k)$ family, supported on the interval $[-\frac{d}{2}, \frac{d}{2}]$.

4. Denote the family of mean-0 *Sinusoidal distributions* as $\text{Sinu}_0(d, s, k)$. Note that d alone dictates the support of the distribution. If $\mu = \mathbb{E}(X|X \sim \text{Sinu}(s, k))$ of a Standard Sinusoidal distribution, then $\text{Sinu}_0(d, s, k) \equiv \text{Sinu}(-\mu d, d, s, k)$. This is an important construct for linear regression setups.

5. s (or any transformation of it) does not affect the shape of the distribution symmetrically around the point $s = 1$; i.e. $\nexists s' < 1 : \text{Sinu}(s', k) \overset{d}{=} 1 - \text{Sinu}(s, k) \quad \forall s > 1$. It is for this reason that we introduced the notion of a flipped *Sinusoidal distribution*.

6. k affects kurtosis non-monotonically. See Figure 2.2.

7. s, k jointly affect both the skewness and the kurtosis. See Figure 2.2.

8. A $\text{Unif}(a, a + d)$ density results from the *Sinusoidal distribution* density when k=0 or s=0.

9. Degenerate at a for $d = 0$

10. $\text{Sinu}(-\frac{d}{2}, d, 1, k) \xrightarrow[d=\pi\sqrt{k}]{k\to\infty} \mathcal{N}(0, 1)$. See Theorem 2.3.1.

11. The mode of $\text{Sinu}(s, k)$ is $2^{-1/s}$ and the modal density is $1/A(s, k)$. See Theorem B.0.1.

2.2 Preliminary Mathematical Properties

Proofs to theorems mentioned are provided in the Appendix section B.1.

2.2.1 Physical Interpretation

Similar to Gilbert 1893 – which imagines intensity (interpreted as probability density) of crater activity on a circular planet as proportional to Gilbert's Sine density $\sin\theta$ – the Standard *Sinusoidal distribution* $\mathrm{Sinu}(s,k)$ can also be physically interpreted as the intensity of crater activity on a (possibly non-circular) planetary surface given by the radial curve:

$$r(z) = r_0 \exp\left(\int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t\right) \tag{2.5}$$

where the curve $r(z)$ is parametrized by $z \in [0,1]$ such that $\theta = \pi z^s$ is the angle the location $r(z)$ subtends at the origin, and r_0 is the initial radius at $z = 0$. Details are furnished in Remark B.1.1.

Theorem 2.2.1 (Joint density expression)**.** *For* $X_1 \ldots X_n \overset{iid}{\sim} \mathrm{Sinu}(a,d,s,k)$*, their joint density can be expressed as:*

$$f(x_1 \ldots x_n) = \frac{1}{A^n(s,k)2^{nk}d^n}\left(\sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1}\left[\prod_{j=1}^n r_j\right]\cdot\exp\left[\sum_{j=0}^n \mathrm{i}\pi r_j\left(\frac{x_j-a}{d}\right)^s\right]\right)^k \tag{2.6}$$

2.2.2 Concentration Inequalities

Theorem 2.2.2 (*Sinusoidal distribution* is sub-Gaussian)**.** $X \sim \mathrm{Sinu}(a,d,s,k)$ *is sub-Gaussian with parameter* $\frac{d^2}{4}$*.*

Proof. $\because a \le X \le a+d \quad \text{a.s.}, \therefore \mathbb{E}\left[\mathrm{e}^{\lambda(X-\mathbb{E}X)}\right] \le \exp\left(-\frac{\lambda d^2}{8}\right)$ by Hoeffding's lemma.

Note that because it is sub-Gaussian, it is also sub-Exponential. □

2.2.3 Normalizing constant for the symmetric case

Theorem 2.2.3 (Normalizing constant, symmetric case)**.**

$$A(1,k) \equiv \int_0^1 \sin^k(\pi t)\mathrm{d}t = \frac{2^k}{\pi}\frac{\Gamma^2\left(\frac{k+1}{2}\right)}{\Gamma(k+1)} \tag{2.7}$$

$$= \frac{1}{\sqrt{\pi}}\frac{\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)} \tag{2.8}$$

2.2.4 Incomplete integral of the kernel for the Symmetric case

Theorem 2.2.4 (Incomplete integral of the kernel for the symmetric case)**.** *When $k \in \mathbb{Z}^+$, $s = 1$, we have –*

$$\int_0^z \sin^k \pi t \, dt = \frac{1}{2^{2n}} \binom{2n}{n} z + \frac{(-1)^n}{\pi 2^{2n-1}} \sum_{k=0}^{n-1} (-1)^k \binom{2n}{k} \frac{\sin 2(n-k)\pi z}{2(n-k)} \qquad [k \text{ even}] \tag{2.9}$$

$$= \frac{1}{\pi 2^{2n}} (-1)^n \sum_{k=0}^{n} (-1)^k \binom{2n+1}{k} \frac{1 - \cos(2n+1-2k)\pi z}{2n+1-2k} \qquad [k \text{ odd}] \tag{2.10}$$

2.3 Analytical replication

We want to investigate parametric limiting cases of the *Sinusoidal distribution* and assess the strength of such convergences. For example, it would be a task to investigate if the *Sinusoidal distribution* converges uniformly to the Normal, Beta, Gamma, etc. distributions in limits of its parameters. In this domain, we have discovered 1 result till now.

2.3.1 Limiting Gaussianity

Theorem 2.3.1. *For the symmetric about 0* Sinusoidal distribution Sinu $\left(-\frac{d}{2}, d, 1, k\right)$, *i.e. with $a = -\frac{d}{2}$ and $s = 1$, setting $d \equiv d(k) = \pi\sqrt{k}$ and letting $k \uparrow \infty$, we have:*

$$\psi_k(x) \to \phi(x)$$

where $\phi(x)$ is the PDF of a Standard Normal distribution.

2.4 Properties involving the Generalized Hypergeometric Function

For this section, we define Generalized Hypergeometric Function (GHGF) F_q^p, parametrized by $a_1, \ldots, a_p$ and $b_1, \ldots, b_p$, calculated at the argument x as:

$$\mathrm{F}_q^p \begin{pmatrix} a_1 & \cdots & a_p \\ b_1 & \cdots & b_q \end{pmatrix}; x \Bigg) := \sum_{j=0}^{\infty} \frac{(a_1)^{\overline{j}} \cdots (a_p)^{\overline{j}}}{(b_1)^{\overline{j}} \cdots (b_q)^{\overline{j}}} \frac{x^j}{j!}$$

where $(x)^{\overline{n}} := x(x+1)\cdots(x+n-1)$ is the rising factorial / Pochhammer function.

Also define from hereon $\binom{r}{k} = \frac{(r)^{\overline{k}}}{k!}$ as the generalized binomial coefficient.

Some additional notes are provided in Remark B.2.1. Proofs to the theorems mentioned are provided in section B.2.

2.4.1 Kernel

Theorem 2.4.1 (Kernel expression via Binomial expansion of Euler form).

$$\varsigma(z; s, k) \equiv \sin^k(\pi z^s) = \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \mathrm{e}^{\mathrm{i}(k-2j)\pi z^s} \tag{2.11}$$

Theorem 2.4.2 (Kernel expression via Generalized Hypergeometric Function).

$$\varsigma(z; s, k) \equiv \sin^k(\pi z^s) = (\pi z^s)^k \left(\mathrm{F}_1^0 \left[\begin{matrix} \cdot \\ \frac{3}{2} \end{matrix} ; -\frac{\pi^2 u^2}{4} \right] \right)^k \tag{2.12}$$

$$= \mathrm{F}_0^1 \left[\begin{matrix} k \\ \cdot \end{matrix} ; -\exp(-2\mathrm{i}\pi z^s) \right] \tag{2.13}$$

2.4.2 CF and MGF

Theorem 2.4.3 (Characteristic Function and MGF of the Standard *Sinusoidal distribution*). *For $Z \sim \mathrm{Sinu}(s, k)$, its CF is given as –*

$$\varphi_Z(t) = \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{m=0}^{\infty} \frac{(\mathrm{i}t)^m}{(m+1)!} \mathrm{F}_1^1 \left[\begin{matrix} \frac{m+1}{s} \\ \frac{m+1}{s} + 1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right] \tag{2.14}$$

Additionally, the MGF is given as –

$$\phi_Z(t) \equiv \varphi_Z(-it) = \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{m=0}^{\infty} \frac{t^m}{(m+1)!} \mathrm{F}_1^1 \left[\begin{matrix} \frac{m+1}{s} \\ \frac{m+1}{s} + 1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right] \tag{2.15}$$

2.4.3 Moments

Theorem 2.4.4 (Moments of Standard *Sinusoidal distribution*). *For $Z \sim \mathrm{Sinu}(s, k)$, $\mu_r(s, k) := \mathbb{E}[Z^r]$ is given by*

$$\mu_r(s,k) = \frac{1}{A(s,k)(2\mathrm{i})^k (r+1)} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \mathrm{F}_1^1 \left[\begin{matrix} \frac{r+1}{s} \\ \frac{r+1}{s} + 1 \end{matrix} ; \mathrm{i}\pi(k-2j) \right] \tag{2.16}$$

$$= \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} \binom{k}{j} (-1)^j \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+r+1)} \tag{2.17}$$

2.4.4 CDF

Theorem 2.4.5 (CDF of the Standard *Sinusoidal distribution*).

$$\Psi(z; s, k) = \frac{1}{A(s,k)(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)} z^{ns+1} \tag{2.18}$$

2.4.5 Normalizing constant

Theorem 2.4.6 (Normalizing constant for the Standard *Sinusoidal distribution*)**.**

$$A(s,k) = \frac{1}{(2\mathrm{i})^k} \sum_{j=0}^{\infty} (-1)^j \binom{k}{j} \sum_{n=0}^{\infty} \frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)}$$

2.5 Numerical and Empirical Properties

2.5.1 Approximation of Sum of 2 IID RVs

Although the convolution of two iid Sinusoidal RVs is not theoretically a *Sinusoidal distribution*, we found that some candidate *Sinusoidal distribution* can often provide good approximation to the same. We may call this phenomenon **approximate iid stability**. This is an intuitive desideratum in a linear modelling context, where we would want the sum of residuals to (at least approximately) lie in the same distribution family as the individual residuals.

The following computational study is conducted in this regard. Each convolution of the same 2 $\mathrm{Sinu}(s,k)$ densities over grid of $s,k \in (0,2]$ is attempted to be approximated by some candidate $\mathrm{Sinu}(0,2,s^*,k^*)$, by the mode matching algorithm specified in section 3.2. The reason for choosing $a=0, d=2$ follows heuristically: we expand the support so as to at least contain all values of the sum of the RVs. Hence we attempt the following approximation:

$$\mathrm{Sinu}(s,k) * \mathrm{Sinu}(s,k) \approx \mathrm{Sinu}(0,2,s^*,k^*) \tag{2.19}$$

for some optimal s^* and k^*.

The results of this approximation routine are presented in the form of a heatmap in Figure 2.3. A poor fit (larger Hellinger distance) appears redder in the plot, whereas a good fit (smaller Hellinger distance) appears greener. Note that the scale is negative logarithmic, i.e. it plots the values of $-\log_{10}(\text{Hellinger distance})$ in colour. Information about the best fitting s^*, k^* are omitted from the plot.

2.5.2 Skewness-Kurtosis cover

Choosing the standard $\mathrm{Sinu}(s,k)$ family, we analyze its skewness-kurtosis cover in Figure 2.4 over a finite grid of parameter choices: $s,k = 0.05\,(0.05)\,6$. The plot is colorized to indicate the choice of parameters as denoted in the legend aside.

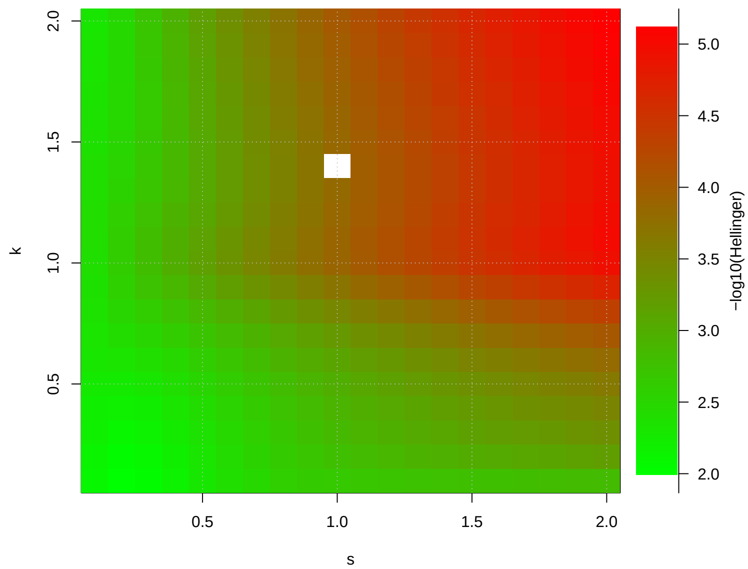


Figure 2.3: Different combinations of (s, k) and the degree of approximability (measured wrt. Hellinger distance) of the self-convoluted $\text{Sinu}(s, k)$ density. A greener tone indicates better approximability. The whitened-out pixels came from garbage values (negative, due to computational limitations), probably because such distributions were completely stable!

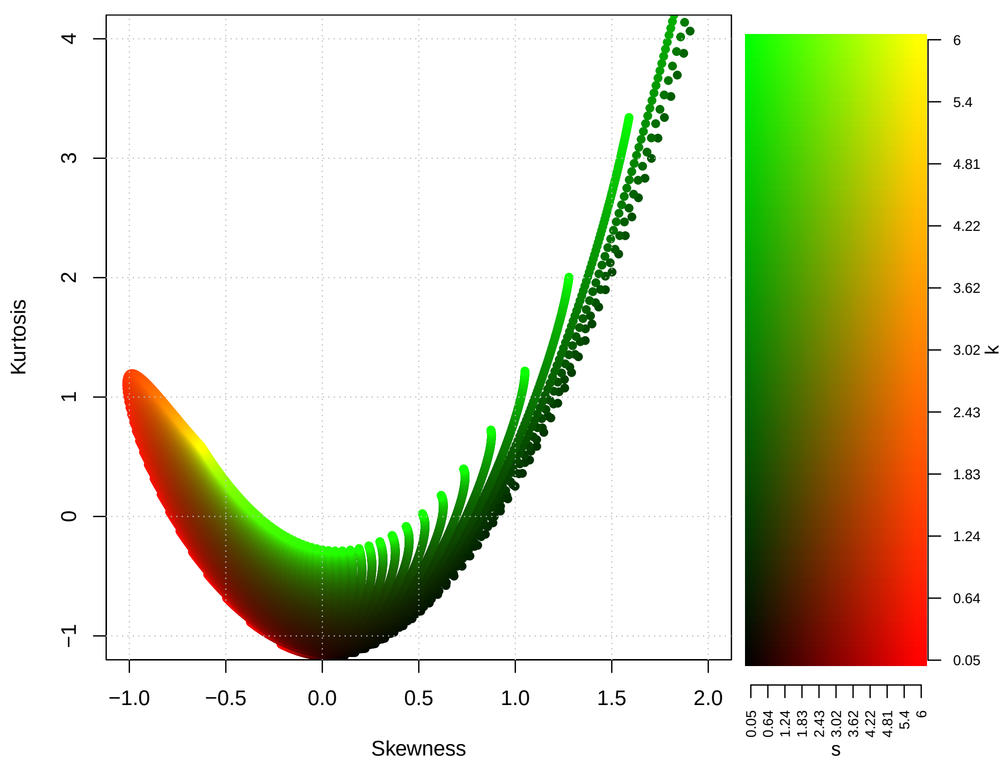


Figure 2.4: Skewness-Kurtosis cover of the *Sinusoidal distribution*.

Chapter 3

Fitting strategies

Firstly, we discuss the feasibility of capturing a distribution via the *Sinusoidal distribution* – i.e. we comment on the goodness of fit. Then we inspect the process of fitting the *Sinusoidal distribution* over multiple kinds of statistical objects – Observational Likelihoods, Modal specifications, MSMs (moment summary measures), ECDFs and Discrete CDFs, and PDFs – in an attempt to demonstrate its flexibility.

3.1 MLE, Score equations

The score equations for maximum likelihood estimation of the parameters of the *Sinusoidal distribution* $\mathrm{Sinu}(a, d, s, k)$ are not analytically solvable, as we detail below. Consider $X_i \overset{\text{iid}}{\sim} \mathrm{Sinu}(a, d, s, k)$, $i = 1(1)n$. The likelihood will be given by

$$L(a, d, s, k; \underset{\sim}{x}) = \prod_{i=1}^{n} \mathbb{1}\{a < x_i < a + d\} \cdot \psi\left(\frac{x_i - a}{d}; s, k\right)$$

Putting the density expression and taking a logarithm, we get the joint log-likelihood as:

$$l(a, d, s, k; \underset{\sim}{x}) = \ln \mathbb{1}\{x_{(1)} > a,\, x_{(n)} > a + d\} - n\ln(d) - n\ln(A(s, k)) + k\sum_{i=1}^{n} \ln\sin\pi\left(\frac{x_i - a}{d}\right)^s$$

Denote $z_i = \frac{x_i - a}{d}$. Some preliminary calculation:

$$\nabla_s A = \pi k \int_0^1 \sin^{k-1}(\pi z^s)\cos(\pi z^s)(\ln z) z^s \; \mathrm{d}z$$
$$\nabla_k A = \int_0^1 \sin^{k}(\pi z^s)\ln\sin(\pi z^s) \; \mathrm{d}z$$

Gives us the score equations:

$$\nabla_a \ell = 0 \iff \sum_{i=1}^{n} z_i^{s-1} \cot(\pi z_i^s) = 0; \qquad \nabla_d \ell = 0 \iff \pi s k \sum_{i=1}^{n} z_i^s \cot(\pi z_i^s) = -n$$

$$\nabla_s \ell = 0 \iff \pi k \sum_{i=1}^{n} \frac{z^s \cos \pi z^s \ln z}{\sin \pi z^s} = n \frac{\nabla_s A}{A}; \qquad \nabla_k \ell = 0 \iff \sum_{i=1}^{n} \ln \sin \pi z_i^s = n \frac{\nabla_k A}{A}$$

which may not be solved analytically. It may be wise to abandon this approach in favour of maximizing the joint likelihood through numerical computation means.

3.2 Mode matching

This method assumes that our target object has a unique, known mode and modal density, and that the support of the candidate *Sinusoidal distribution* is known or soundly estimable. WLoG, assume density is supported on $(0, 1)$. Then, the remaining problem is only to optimize s and k. Observe that s, k jointly control the mode. So "mode matching", described as follows, presents one alternative to estimating s, k:

1. Recall that the mode and the modal density of $\mathrm{Sinu}(s, k)$ are $2^{-1/s}$ and $1/A(s, k)$ respectively.
2. Equate $2^{-1/s}$ = target mode. Call the solution s^*, tractable by analytical calculations alone.
3. Equate $A(s^*, k)$ = target modal density. Call the solution k^*, derivable by root-finding or optimization algorithms.
4. Optimal candidate density is thus $\mathrm{Sinu}(s^*, k^*)$.

It is to be noted that if this optimal density is a poor fit to the target, then either the target is not a Sinusoidal shape, or Sinusoidal distribution approximates is poorly within those bounds.

An example of usage of this method is demonstrated in subsection 2.5.1.

3.3 MSM on MSM

Given a set of 4 moment summary measures (MSMs), we focus on the problem of fitting to it an appropriate *Sinusoidal distribution* of 4 parameters, so that its MSMs match the specifications as closely as possible. For a given set of moment summary measures (MSMs) $\vec{\mu}_0$, the MSM method hence minimizes w.r.t. a, d, s, k the distance L between the target MSMs and those of $\mathrm{Sinu}(a, d, s, k)$:

$$\min_{a,d,s,k} L(\vec{\mu}_0 - \vec{\mu}(a, d, s, k))$$

with a natural choice for L being $\sum_{r=1}^{4} (\mu_r^0 - \mu_r(a, d, s, k))^2$.

Because $\mathrm{Sinu}(a, d, s, k)$ is a location-scale family in (a, d), its first two MSMs (mean & variance) can be freely adjusted by modifying (a, d) alone, it should be sufficient to investigate the skewness-kurtosis adaptability of the distribution through its parameters s and k. A numerical description of the skewness-kurtosis cover of the *Sinusoidal distribution*is provided in Figure 2.4. Our optimization algorithm makes use of this fact by optimizing over s, k jointly first, and then adjusting a, d to meet the desired mean-variance requirements. A demonstration of this approach is presented in Figure 3.1.

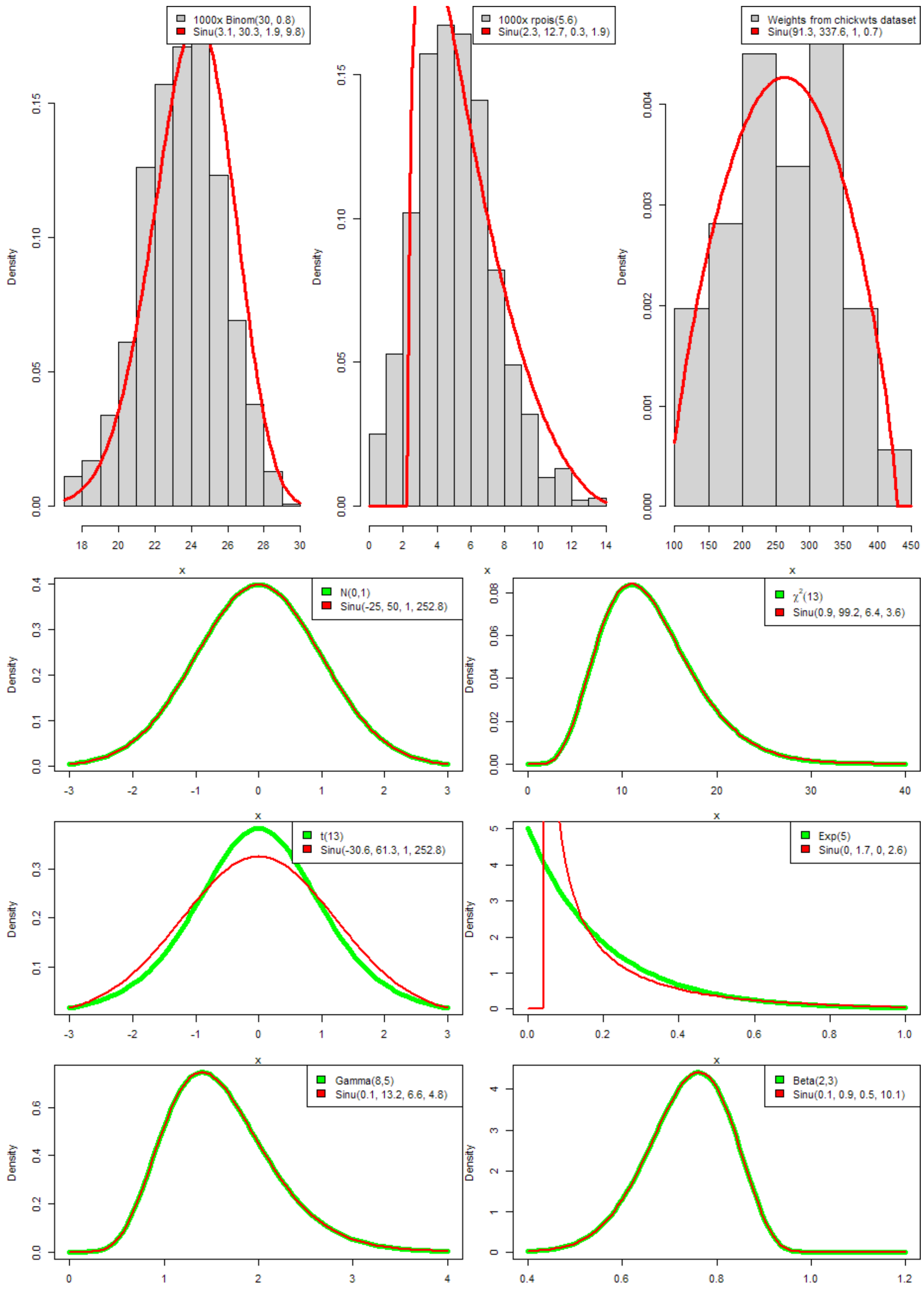


Figure 3.1: MSMs of Sinu(a, d, s, k) fitted over some theoretical and data MSMs.

3.4 PDF on PDF

We tried fitting a Sinu(a, d, s, k) density over different target PDFs (standard location-scale members only) using numerical optimization (`L-BFGS-B` method of the `optim` function found in the `R` programming language) of a PDF-PDF divergence as follows:

$$\min_{a,d,s,k} L(f, g)$$

with a natural choice for L being any density-density divergence metric, for example Hellinger metric $L(f, g) := \int_{\mathscr{X}} \frac{1}{2}(\sqrt{f} - \sqrt{g})^2 \mathrm{d}x$. For finiteness of computation, we operated under the following parametric constraints whilst choosing the best candidate Sinu(a, d, s, k): $a \in [-10^3, 10^3]$, $d \in [10^{-2},\ 2 \times 10^3]$, $s \in [10^{-2}, 10^2]$ and $k \in [10^{-2}, 10^2]$. The results (minimized Hellinger distance with the choice of parameters) are provided in Table 3.1.

Target Density	Hellinger distance (Choice of Parameters a, d, s, k)
$\mathcal{N}(0, 1)$	0.00004277289 (-20.542932, 34.985498, 1.304535, 100)
Lognormal$(0, 1)$	0.000936801 (0.02827762, 1000, 0.08566970, 35.78972)
Skew$\mathcal{N}_{\alpha=1}(0, 1)$	2.189446e-05 (-7.2216216, 24.8516728, 0.5936823, 100)
Logistic$(0, 1)$	0.003149183 (-80.275639, 102.222513, 2.882431, 100)
Exp(1)	0.008101119 (0.0005708485, 6.5585684475, 0.0621246472, 1.6766166256)
DE$(0, 1)$	0.01599712 (-8.895071, 16.872288, 1.085768, 15.608238)
$\Gamma(0.1, 1)$	Poor fit
$\Gamma(100, 1)$	Poor fit
B$(1, 1)$	0.0030202, (-0.0007780335, 1.0012066335, 0.1004892336, 0.12190192670)
Cauchy$(0, 1)$	0.06878475 (-126.211809, 149.303341, 4.141315, 100)
$\chi^2(1)$	Poor fit
$\chi^2(10)$	5.791378e-06 (-0.3201950, 97.1337699, 0.2815837, 22.0271292)
$\mathrm{t}_1(0, 1)$	0.06878477 (-126.114092, 149.207078, 4.138119, 100)
$\mathrm{t}_{10}(0, 1)$	Fit failed
F$(1, 1)$	Poor fit

Table 3.1: Some common target densities and the best candidate *Sinusoidal distribution*

The technical and theoretical underpinnings behind poor or failed fits will be the matter of future investigation.

3.5 CDF on ECDF

We focus on the task of fitting a *Sinusoidal distribution* on a target ECDF G_n by minimizing an L_2 loss of the following form:

$$\min_{a,d,s,k \in \Theta} L(F, G_n)^2$$

with a natural choice for L being $\sum_{x \in X_i} (F - G_n)^2(x)$, where the summation is only over the observed datapoints. We choose such a metric primarily for its general finiteness and differentiability. Note that such techniques may seamlessly be used for discrete CDFs as well, by taking the summation over its masspoints instead.

The results of fitting some common CDFs/ECDFs are given in Figure 3.2.

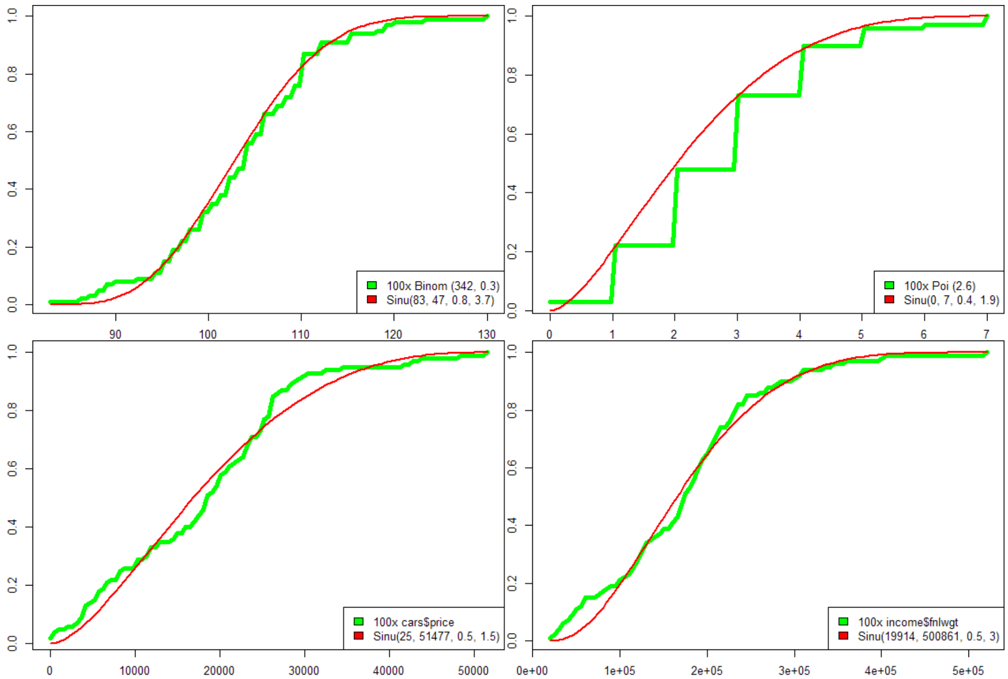


Figure 3.2: CDF of Sinu(a, d, s, k) fitted over some theoretical and data ECDFs.

3.6 Additional Discussion

3.6.1 Testing Sinusoidality of a sample

Perhps before even going into the task of fitting a *Sinusoidal distribution* to a statistical object, given a sample we should check its goodness of fit to the *Sinusoidal distribution* family. This is possible by usual nonparametric procedures such as the Kolmogorov-Smirnov or the Anderson-Darling test. Like in the case of testing for normality, the algorithm involves first estimating the parameters of the "best-fitting" *Sinusoidal distribution* by optimizing an appropriate criterion, such as the joint likelihood. Next, an appropriate non-parametric test is carried out between the fitted density and the observed sample.

In this regard, it is to be noted that the Chi-squared test, which involves binning the observations, before comparing observed and expected frequencies, adds an additional layer of opacity between the sample and the distribution function, by dint of an arbitrary choice of binning. On the other hand, a test like the Shapiro-Wilk test, which depends on specific esoteric properties of the Normal distribution, could not be devised for the *Sinusoidal distribution* as of yet, because no particular identifying property of that sort that can be leveraged similarly has been found yet.

3.6.2 Density-quantile domain divergences

Following Staudte 2017, we may construct the notion of divergences based on the density-quantile function $dq(p) = f(F^{-1}(p)) \quad \forall p \in [0,1]$. A benefit of calculating divergence metrics on the density-quantile domain is that densities (and thereby divergences) are location-scale invariant and invariant to "flat" regions in the CDF. However, the computational complexity of evaluating the quantile function of a *Sinusoidal distribution*, and consequent to the same, optimizing with respect to its parameters some functional of the quantile function, is a strong deterrent against this mode of operation.

3.6.3 Divergence potential map as the determining factor behind fits

For some target density g, if its divergence from any candidate density f, $D_g(f) \equiv D(f,g) : \mathscr{F} \times \mathscr{G} \mapsto \mathbb{R}$ can often be represented as an integral:

$$\int_{x\in\mathcal{X}} L(f(x), g(x))\mathrm{d}x$$

We introduce $V(x,y) = L(y, g(x))$ as the **divergence density/potential associated with** g **and induced by** L. It gives a glimpse at the topography of penalization of $D(\cdot,\cdot)$ with respect to g. Some common divergences with their associated potentials are provided in Table 3.2. Figure 3.3 gives a visualization of the potential associated with a contaminated density.

Divergence name	Form	Potential $L_g(y)$	Gradient $\nabla_y L_g(y)$
Hellinger	$\int \frac{1}{2}(\sqrt{y(x)} - \sqrt{g(x)})^2 \mathrm{d}x$	$\frac{1}{2}(\sqrt{y} - \sqrt{g})^2$	$\frac{1}{2\sqrt{y}}(\sqrt{y} - \sqrt{g})$
DPD-α	$\int y^\alpha(y-g) - \frac{1}{\alpha}g(y^\alpha - g^\alpha)\mathrm{d}x$	$y^\alpha(y-g) - \frac{1}{\alpha}g(y^\alpha - g^\alpha)$	$(1+\alpha)y^{\alpha-1}(y-g)$
Kullback–Leibler	$\int y \log\left(\frac{y}{g}\right)\mathrm{d}x$	$y\log\left(\frac{y}{g}\right)$	$\log\left(\frac{y}{g}\right) + 1$
Reverse KL	$\int g \log\left(\frac{g}{y}\right)\mathrm{d}x$	$g\log\left(\frac{g}{y}\right)$	$-\frac{g}{y}$
Pearson χ^2	$\int \frac{(y-g)^2}{g}\mathrm{d}x$	$\frac{(y-g)^2}{g}$	$\frac{2(y-g)}{g}$
Neyman χ^2	$\int \frac{(y-g)^2}{y}\mathrm{d}x$	$\frac{(y-g)^2}{y}$	$1 - \frac{g^2}{y^2}$
Total variation	$\int \lvert y-g\rvert\mathrm{d}x$	$\lvert y-g\rvert$	$\mathrm{sign}(y-g)$
Jensen–Shannon	$\frac{1}{2}\int\left[y\log\left(\frac{2y}{y+g}\right) + g\log\left(\frac{2g}{y+g}\right)\right]\mathrm{d}x$	$\frac{1}{2}y\log\left(\frac{2y}{y+g}\right) + \frac{1}{2}g\log\left(\frac{2g}{y+g}\right)$	$\frac{1}{2}\log\left(\frac{2y}{y+g}\right)$

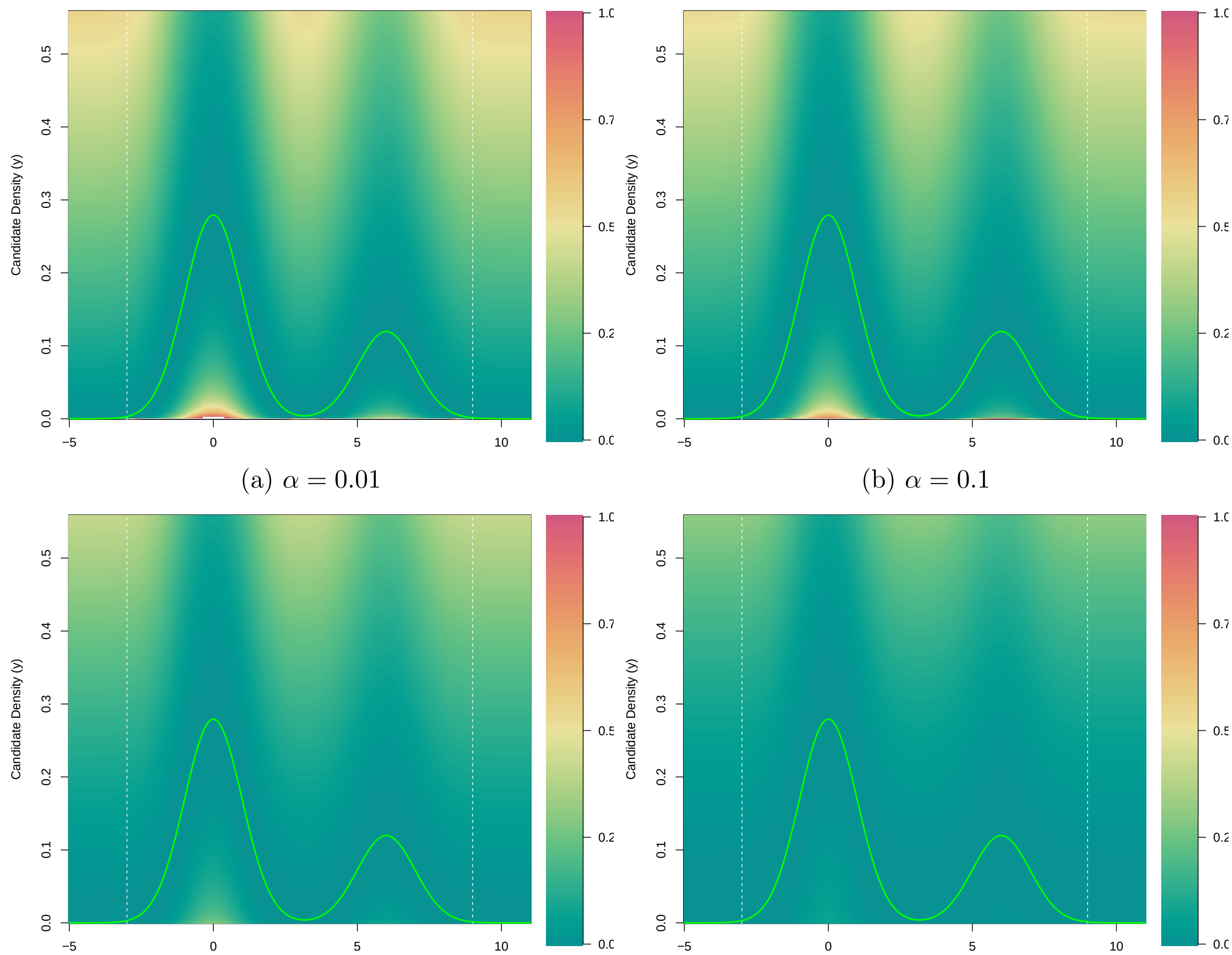
Table 3.2: Some common PDF-PDF divergences and their potential functions

Note how almost all divergences are functions of $f - g$, i.e. dependent on the abscissa-wise deviation of the candidate density from the target. However the rate of increment of penalization wrt. the deviation changes with the divergences.

The process of fitting a candidate distribution over a given target then becomes merely an act of minimizing the integral of potential along the density-line, with the constraint of area under the density line as 1.

DPD and contamination

To understand the influence of the tuning parameter α on the DPD metric, with the ambition of suggesting novel ways of estimating the same, a long-standing research question (Basak, Basu, and Jones 2021), we observe heuristically that a higher α causes a lower rate of increase of penalization for deviating from true curve, thereby providing more feasibility of a fit that covers both density portions. Whereas a lower α allows rigid maximization over one of the 2 curves only, since passing through the middle high-penalization region in between incurs a lot of cost. See Figure 3.3.



(a) $\alpha = 0.01$ (b) $\alpha = 0.1$

(c) $\alpha = 0.5$ (d) $\alpha = 1$

Figure 3.3: Divergence potentials induced by different α's of the contaminated density $0.7\mathcal{N}(0,1) + 0.3\mathcal{N}(0,1)$

We find that for contaminated densities, it is not the "attraction" of the second peak that deviates the candidate to fit contaminatedly, but it is the infeasibility in between that makes unimodal densities suboptimal fits, thereby seeking that the candidate choose either the principal or the contaminant density component.

Chapter 4

Applications

4.1 Unconventional datasets

Whilst classical procedures bank heavily upon either the assumption of Gaussianity in the exact sample domain, or employ Gaussianity results in the large sample domain, nonparametric procedures come up with distribution-invariant results with great generalization at the cost of reduced power and confidence. Still, both approaches struggle with phenomena such as bounded support, asymmetric and kurtotic concentration, non-polynomial tail decay, and multimodal tendencies. In many applications, forcing such datasets into Gaussian or exponentially-tailed frameworks may lead to inaccurate uncertainty quantification, unstable inference, and poor finite-sample approximations.

We aim to strike a middle ground of balance, where the shape-flexibility of our *Sinusoidal distribution* leads to more flexible data-distributional and test-statistic distributional models. This aids in the analysis of unconventional, specifically skewed or platykurtic, datasets. Such data-fitting exercise will constitute the foundational basis for later procedures like testing and inference.

An example of fitting the *Sinusoidal distribution* to data was demonstrated in section 3.3 and section 3.5.

4.2 Test statistic distribution modelling

Consider the elementary classical hypothesis testing problem $H_0 : \mu = 0$ vs. $H1 :$ not H_0, based on observed sample $X_1, X_2, \dots, X_n \sim F$ with $\mathbb{E}_F X_i = \mu$. The general assumption during the formulation of the convenient one-sample Student's t-test is that $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Under this assumption, the statistic $T_n = \frac{\sqrt{n}\bar{X}}{S} \sim \mathrm{t}_{n-1}$ under H_0.

However, when the underlying distribution is significantly non-normal, the finite-sample distribution of T_n may differ substantially from a t-distribution, thereby resulting in inaccurate p-values if t-testing is forcefully conducted nevertheless.

To address this issue, we propose a sinusoidal approximation strategy based on the *Sinusoidal distribution* family. Because of its highly flexible nature, the *Sinusoidal distribution* is expected to serve as a good proxy distribution for not only complicated distributions of usual test-statistics, but also distributions of test-statistics of complicated forms themselves. Hence, inference can be done accurately against the approximating *Sinusoidal distribution*, thereby eliminating the need for elaborate tables in all practical purposes.

4.2.1 Methodology and results

Suppose that the underlying observations arise from a *Sinusoidal distribution* $X_i \overset{\text{iid}}{\sim} \text{Sinu}(a, d, s, k)$, where the parameters d, s, k are estimated from the observed sample using any of the methods described in Chapter 3. Since $H_0 : \mu = 0$, a is chosen such that $\mathbb{E}[X_i] = 0$.

We then approximate the H_0 distribution of the test statistic via Monte Carlo simulation for B (say =100) iterations as follows:

1. Generate independent Monte Carlo samples $X_1^{(b)}, \ldots, X_n^{(b)} \overset{\text{iid}}{\sim} \text{Sinu}(a, d, s, k), \qquad b = 1, 2, \ldots, B.$
2. For each simulated sample, compute the statistic $T_n^{(b)} = \frac{\sqrt{n}\,\bar{X}^{(b)}}{S^{(b)}}$.
3. Collect the simulated values $T_n^{(1)}, \ldots, T_n^{(B)}$ to obtain an empirical approximation to the null distribution of the statistic.
4. Fit another *Sinusoidal distribution*

$$\text{Sinu}(a_T, d_T, s_T, k_T)$$

   to the simulated statistic values $T_n^{(1)}, \ldots, T_n^{(B)}$.
5. Use the fitted *Sinusoidal distribution* to compute p-values.

We employed this approach to model the t-statistic distribution of 50 observations from a $\text{Sinu}(a_0, 2, 3, 4)$ data distribution (where a_0 ensures mean 0) by 100 Monte-Carlo iterations. The obtained ECDF and the best fitting *Sinusoidal distribution* are illustrated in Figure 4.1.

4.2.2 Discussion

Our procedure empirically replaces the classical Student's-t approximation with a data-adaptive Sinusoidal approximation of finite-sample H_0 distribution of the t-statistic. Unlike the usual t-test, this method does not rely upon exact Normality of the observations.

The approach may be interpreted as a **parametric bootstrap** strategy in which both the underlying data distribution as well as the induced statistic distribution are modelled Sinusoidally.

As the sample size increases, one may expect convergence toward the usual Gaussian asymptotics under suitable regularity conditions. However, in small or moderate samples arising from strongly non-Normal populations, the sinusoidal approximation may provide improved p-values and rejection probabilities.

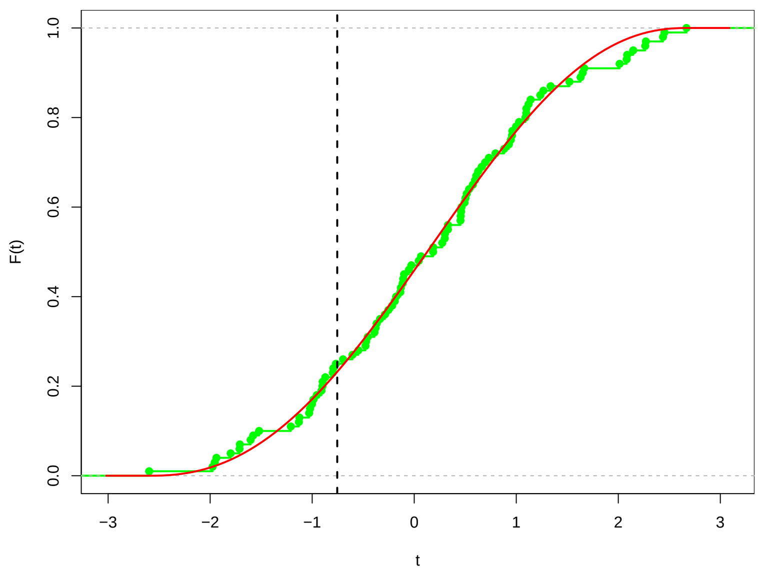


Figure 4.1: MC simulation (green) and a Sinusoidal modelling (red) of the classical Student's t-statistic when underlying data distribution came from an unconventional Sinu$(a_0, 2, 3, 4)$ distribution.

4.3 Sinusoidal error regression

In this section, we test the effectiveness of *Sinusoidal distribution* in a simple linear regression setting by using it as a flexible distribution for the error terms. The linear model will hence be:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

– where $\epsilon_i \overset{\text{iid}}{\sim} \text{Sinu}(a, d, s, k)$ and all model parameters $\beta_0, \beta_1, a, d, s, k$ are unknown – in modelling various datasets. Our methodology will closely follow Dean and King 2009.

We will resort to numerical ML Estimation of model parameters. Observe that for SER, the data loglikelihood is:

$$\begin{aligned}\ln L(\beta_0, \beta_1, a, d, s, k) &\equiv \ln \prod_{i=1}^{n} f(e_i; \beta_0, \beta_1, a, d, s, k) \\ &= \sum_{i=1}^{n} \ln \psi \left(\frac{(y_i - \beta_0 - \beta_1 x_i) - a}{d}; s, k \right)\end{aligned}$$

which gives the MLE of the parameters as $\arg\max_{\beta_0, \beta_1, a, d, s, k} \sum_{i=1}^{n} \ln \psi \left(\frac{(y_i - \beta_0 - \beta_1 x_i) - a}{d}; s, k \right)$. Since analytical solutions are intractable, we will resort to a numerical optimization (L-BFGS-B) routine to extract ML estimates.

4.3.1 Approximating usual Gauss-Markov setting

First, we analyze the performance of *Sinusoidal distribution* in approximating a Gauss-Markov model with standard Normal error. Recollect in this regard that we already know that *Sinusoidal distribution* provides an arbitrarily well approximation of the Normal distribution for parameters in their limiting form. We repeated 50 iterations of the following experiment, with the arbitrary seed `set.seed(1234)` for reproducibility, for each of the sample sizes $n = 50, 100, 150, 200$ and collected performance metrics:

1. Fix a sample size, say $n = 50$.
2. Generate 2 arbitrary coefficients $\beta_0 \sim U(10, 20)$ and $\beta_1 \sim U(5, 10)$, and a data column $x_i \sim U(0, 1)$, $i = 1(1)n$. These stay fixed across all iterations.
3. For every new iteration:
   (a) Simulate observational errors $\epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$, $i = 1(1)n$. Simulate the response as $y_i = \beta_0 + \beta_1 * x_i + \epsilon$.
   (b) Conduct Normal error regression, which is simply OLS regression. Note the coefficients.
   (c) Conduct SER via numerical loglikelihood maximization. Note the coefficients.
4. After all iterations are over, compare the NER coefficients against the SER coefficients.
5. Move to the next sample size.

Our findings are described in Table 4.1. Note that we obtained true $\beta_0 = 11.137$ and $\beta_1 = 8.111$ respectively.

Sample Size	OLS $\hat{\beta}_0$ Mean (Variance)	$SER \hat{\beta}_0$ Mean (Variance)	OLS $\hat{\beta}_1$ Mean (Variance)	SER $\hat{\beta}_1$ Mean (Variance)
10	11.175 (1.006)	11.163 (1.024)	7.970 (3.000)	8.031 (3.208)
25	11.103 (0.234)	11.103 (0.234)	8.219 (0.694)	8.208 (0.690)
50	11.363 (1.227)	11.210 (0.110)	8.008 (0.619)	7.960 (00.549)
100	11.155 (0.034)	11.153 (.034)	8.053 (0.144)	8.069 (0.139)
150	11.189 (0.019)	11.191 (0.018)	8.013 (0.067)	8.009 (0.066)

Table 4.1: Performance of SER against NER when errors were actually normally distributed

We observe that due to severe computational complexities as described in subsection 4.3.2, there is no significant additional gain by implementing SER in a Gauss-Markov model. In fact, the estimate variances for a, d, s, k are also unusably large for any practical or theoretical appreciation.

4.3.2 Discussion and Extensions

The optimization routine for likelihood estimation in a regression setup with a bounded support error-distribution such as the *Sinusoidal distribution* is particularly difficult. It contains, beyond the usual complexity of constrained maximization, the additional complexity of ensuring support coverage of the residuals at all times. i.e., for every β_0, β_1 chosen by the optimizer, a new set of estimates for the residuals are obtained, and consequently the optimizer's choice of a, d must ensure that all errors lie within the support $[a, a+d]$. Hence the problem is manifold: where the region of constrained optimization itself varies with the optimizer's trajectory.

As a result of this, more complicated models such as SER on actually Sinusoidally distributed errors, and on arbitrary datasets with no distributional specification, are harder to implement. Safer, heuristic approaches such as modelling OLS residuals and choosing large enough supports for the error distribution, may however be tried.

4.4 Flexible Activation Function for NNs

We have already seen in Table 3.1 that the *Sinusoidal distribution* can take the shape of a Logistic (0,1) distribution with great accuracy. Given the closeness of its CDF to the sigmoid activation function, and the additional ability to take on other shapes as well, we aim to construct a neural network endowed with a flexible skewness-kurtosis activation function. This learnable-parameter NN will require deriving custom back-propagation equations for parameter estimation. We will also assess the additional cost of training such a neural network and the benefits associated with the same.

Buzsáki and Mizuseki 2014 highlights the presence of skewed activation functions in the brain itself. That asymmetric activation functions perform well has been studied in Júnior et al. 2025. Küçükalİ et al. 2026 have also constructed several new skewed activation functions.

4.4.1 Varieties of activation functions

Demarcate the **bottom (B) and upper (U) "arms"** of an activation function (equivalently its left and right arms respectively, since an activation function is always non-decreasing) as the 2 parts of the activation function's curve close to horizontal but not exactly flat. We choose the portions of the curve contained in the Y-axis windows $(0, 0.1]$ and $[0.9, 1)$ for the purpose of demarcating the arms, although a more objective method of demarcation may involve the inflexion points. Call the segment joining the 2 arms a **ramp**.

	$s = 0.5$	$s = 1$	$s = 3$
$k = 1$	Early onset Early saturation	Moderate onset Early saturation	Delayed onset Delayed saturation
$k = 3$	Early onset Moderate saturation	Moderate onset Moderate saturation	Delayed onset Moderate saturation
$k = 5$	Early onset Delayed saturation	Moderate onset Moderate saturation	Delayed onset Moderate saturation

Table 4.2: A heuristic interpretation of the activation behaviour of various Sinu activation functions based on the lower-arm (B) and upper-arm (U) lengths.

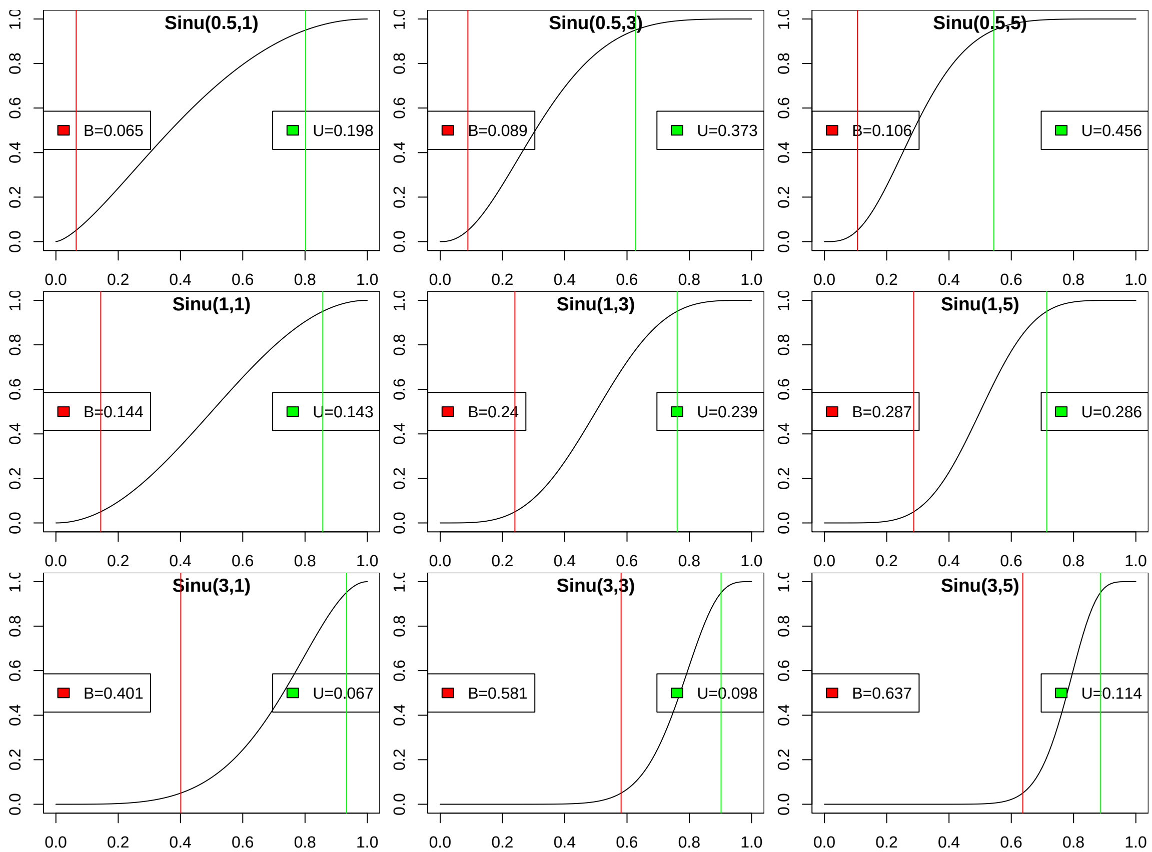


Figure 4.2: Different *Sinusoidal distributions* as activation functions. U and B are the lengths of the upper and bottom arms respectively. Each upper arm starts at the green line and goes upto x=1 and each bottom arm starts at x=0 and goes upto the red line. The segment in between the 2 lines is the "ramp".

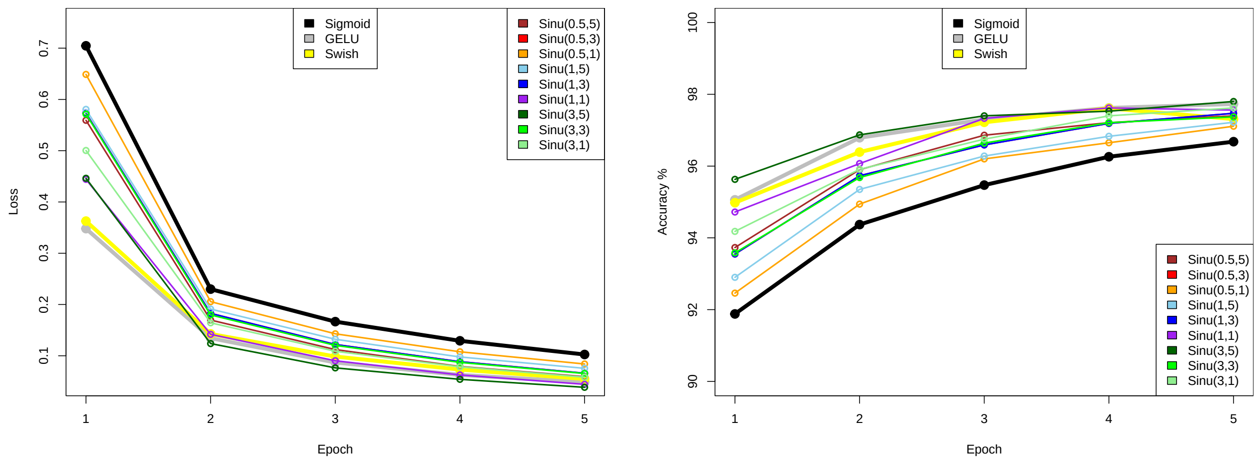
A note about the nomenclature is as follows: Smaller values of B indicate that the activation function leaves the neighbourhood of 0 quickly, corresponding to an *early onset*, whereas larger values of B imply a more *delayed onset*. Similarly, since U denotes the length of the upper arm before attaining the saturated state 1, smaller values of U indicate that the function reaches saturation only near the end of the domain, corresponding to *delayed saturation*. Conversely, larger values of U imply an *early saturation*. Intermediate values are heuristically termed *moderate*.

Note how an s far-away from 1 (close to 0 XOR ∞) causes imbalanced arms. Again, a larger k causes larger arms and a steeper ramp thereby.

4.4.2 Methodology and Results

We chose one of the most elementary machine learning routines – classification of the MNIST dataset – using a relatively simple feed-forward neural network architecture with 2 hidden layers of sizes 256 and 128 respectively. The input layer gets a 28×28 matrix of black-and-white pixels of handwritten digits, i.e. upon normalization, a vector from $[0, 1]^{28\times 28}$. Corresponding to each digit is present a label: the true digit value of the inscription. The model trains on a training subset of the data

To ensure fair comparison of the effectiveness of the aforementioned 9 different standard *Sinusoidal distributions* as activation functions, we replaced them in the aforementioned model each time as the candidate activation function, keeping all other model properties unchanged. Also for reference, we took the (a) vanilla sigmoid, (b) GELU, and (c) SiLU/Swish functions, using them one-by-one as activation functions in the otherwise intact NN architecture. More details about the GELU and SiLU activation functions are available in subsection C.1.1.



(a) Loss of different models

(b) Accuracy of different models

Figure 4.3: Performance of different neural network models over 5 epochs of training over the MNIST dataset.

It can be safely said that all activation functions give a generally usable NN model on this particular dataset. The differences are marginal, all within a quite high accuracy / low loss window.

With respect to the control/reference activation functions, we first observe that the Sigmoid activation NN model underperforms in accuracy and undergoes a slower optimization of the loss function than all its competitors. On the other hand, GELU, an activation function frequently used in transformer based NNs, performs better than most of its competitors. Swish is the industry standard for classification tasks: hence it is no wonder that it maintains top-of-the-line performance.

Coming to the *Sinusoidal distribution*activation function models, the results are varied. The left skewed platykurtic Sinu$(0.5, 1)$ underperforms with the largest loss and the least accuracy amongst its peers by the end of all 5 epochs. The symmetric Sinu$(1, 5)$ and Sinu$(1, 3)$ fare only slightly better. However, most interestingly, the right skewed and leptokurtic Sinu$(3, 5)$ model almost consistently outperforms even the industry standard Swish model in both and accuracy metrics over 4 out of 5 epochs.

4.4.3 Discussion and considerations

An important insight this approach provides us is that **datasets have their own optimal activation functions**. Whether different datasets share the same optimal activation function, and what factors influence optimality, are questions for further investigation. However, a strong case is to be made for learnable-parameter NNs based upon a flexible, parametric family of activation functions. This setup will allow the data to refine the models' flexible activation function freely as per its own optimality requirements.

During computational experimentation, we found that a Sinu($s = 5, k = 8$) model performed very poorly, with a reported loss of 2.3013 and a test accuracy of 11.35% remaining constant over all 5 epochs. It is suspected that a high skewness and high kurtosis produced a steep ramp at the right end, which was virtually inapproximable by our chosen grid approximation and caching methods. Hence the prediction offered by this model was garbage.

Conclusion

In the current iteration, we saw the applicability of the *Sinusoidal distribution* to real-world problems – data fitting, test-statistic modelling, regression and as neural network activation functions. We conclude that the *Sinusoidal distribution* is a highly potent distribution which can find application in a wide variety of statistical tasks once we leverage its flexibility that comes at the cost of its computational complexity.

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Appendix A

Code

All code for this project is available at github.com/BoiBhai/Sinusoidal-Distribution.git.

A.1 Choice of Optimizer

Following optimization techniques were tried and their drawbacks realized:

1. `optim`'s `L-BFGS-B` method: sensitivity on initial choices of parameters.

2. `optim`'s `Nelder-Mead` method: unavailability of bounding boxes and the presence of unbounded solutions for certain problems such as $\mathcal{N}(0, 1)$ fitting.

Appendix B

Proofs and Remarks

Theorem B.0.1. *Mode of* $\mathrm{Sinu}(s,k)$ *is* $2^{-1/s}$ *and the modal density is* $\frac{1}{A(s,k)}$.

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}z}\psi(z;s,k) &= \frac{1}{A(s,k)}\frac{\mathrm{d}}{\mathrm{d}z}\sin^k(\pi z^s) \\
&= \frac{1}{A(s,k)}k\sin^{k-1}(\pi z^s)\cos(\pi z^s)\pi s z^{s-1} \\
&= 0 \iff \begin{cases} \sin(\pi z^s) = 0 \implies z = 0, 1 \\ z^{s-1} = 0 \implies z = 0 \\ \cos(\pi z^s) = 0 \implies \pi z^s = \frac{\pi}{2} \implies z = 2^{-1/s} \end{cases}
\end{aligned}$$

And modal density is

$$\psi(2^{-1/s};s,k) = \frac{1}{A(s,k)}\sin^k(\pi z^s)\bigg|_{z=2^{-1/s}} = \frac{1}{A(s,k)}$$

□

B.1 Preliminary mathematical properties

Remark B.1.1 (Physical Interpretation of the *Sinusoidal distribution*). First, note that both the radius of a planet and the angle subtended at its center are actually immaterial with regard to the intensity of crater activity it experiences when we assume that there is no decay in the intensity of crater activity through space, or atleast in the region of space incident normally upon the planet starting from its north-pole to its equator. In fact, what only matters is the tangential inclination of any location on the planet. To summarise, a location $r(z)$ parametrized as above will experience activity intensity proportional to $\sin(\phi)$, where ϕ is the angle its tangent subtends against the vertical.

Now, consider a planet whose every location $r(z)$ is inclined at an angle $\phi = \sin^{-1}[\sin^k(\pi z^s)]$ against the vertical. it is immediate that the activity intensity there would be $\sin^k(\pi z^s)$.

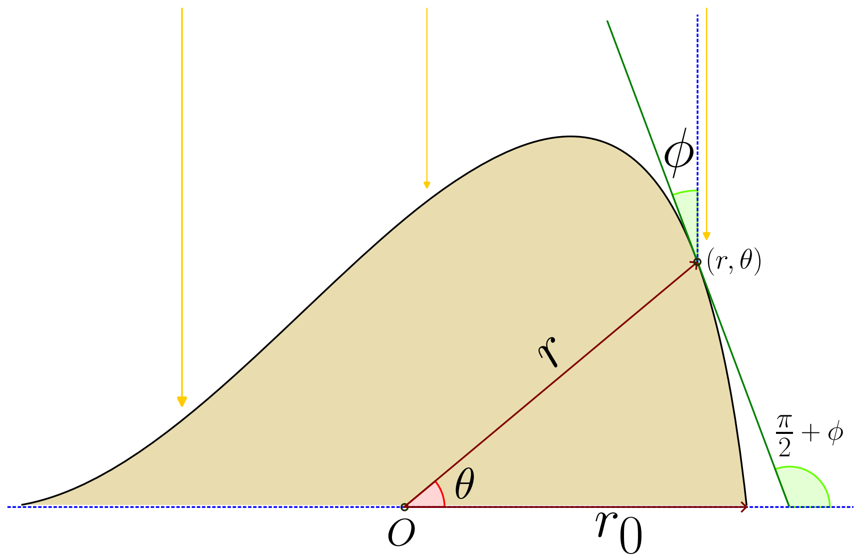

Figure B.1: An interpretation of the *Sinusoidal distribution*

The characterization of the shape of such a (smooth) radial curve that defines the planet is a problem in differential geometry. We briefly formulate its solution:

$$
\begin{gathered}
x(0) = r_0;\ y(0) = 0 \\
\phi(z) = \sin^{-1}[\sin^k(\pi z^s)] \\
\frac{\mathrm{d}y}{\mathrm{d}x} = \tan\left(\frac{\pi}{2} + \phi\right) \equiv -\cot\phi
\end{gathered}
$$

$$
\begin{aligned}
x = r(z)\cos(\pi z^s) \quad &\Longrightarrow\ \mathrm{d}x = \cos(\pi z^s)\mathrm{d}r - r\sin(\pi z^s)\pi s z^{s-1}\mathrm{d}z \\
y = r(z)\sin(\pi z^s) \quad &\Longrightarrow\ \mathrm{d}y = \sin(\pi z^s)\mathrm{d}r + r\cos(\pi z^s)\pi s z^{s-1}\mathrm{d}z
\end{aligned}
$$

$$
\begin{aligned}
\frac{\sin(\pi z^s)\mathrm{d}r + r\cos(\pi z^s)\pi s z^{s-1}\mathrm{d}z}{\cos(\pi z^s)\mathrm{d}r - r\sin(\pi z^s)\pi s z^{s-1}\mathrm{d}z} \qquad &= -\cot\left[\sin^{-1}\sin^k(\pi z^s)\right] \\
\Longrightarrow \frac{\mathrm{d}r}{r} \qquad &= \tan\left[\pi z^s - \sin^{-1}\sin^k(\pi z^s)\right]\cdot \pi s z^{s-1}\mathrm{d}z \\
&= \tan(\theta - \sin^{-1}\sin^k\theta)\mathrm{d}\theta \\
\Longrightarrow \ln r\Big|_{r_0}^{r(z)} \qquad &= \int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t \\
\Longrightarrow r(z) \qquad &= r_0\exp\left(\int_0^{\pi z^s} \tan(t - \sin^{-1}\sin^k t)\mathrm{d}t\right)
\end{aligned}
$$

Proof of ??. Consider the transformation $Y = \sin^2(X)$ for $X \in [0, \pi/2]$. The Jacobian of the transformation is:

$$\left|\frac{dx}{dy}\right| = \frac{1}{2\sin(x)\cos(x)} = \frac{1}{2y^{1/2}(1-y)^{1/2}} \tag{B.1}$$

Substituting the density $f_X(x) \propto \sin^K(x) = y^{K/2}$:

$$f_Y(y) \propto y^{K/2} \cdot y^{-1/2}(1-y)^{-1/2} = y^{\frac{K+1}{2}-1}(1-y)^{\frac{1}{2}-1} \tag{B.2}$$

This identifies Y as Beta $\left(\frac{K+1}{2}, \frac{1}{2}\right)$. The full support $[0, \pi]$ is obtained by reflecting X across $\pi/2$ with probability 0.5 via the Bernoulli variable S.

Let $U = \pi Z^S$. The Jacobian of the inverse transformation $z = (u/\pi)^{1/S}$ is $|dz/du| \propto u^{(1-S)/S}$. The density of U is:

$$f_U(u) \propto f_Z((u/\pi)^{1/S}) \cdot u^{(1-S)/S} \propto (u^{(S-1)/S}\sin^K u) \cdot u^{(1-S)/S} = \sin^K u \tag{B.3}$$

on $u \in [0, \pi]$. By the Beta-transformation theorem for $\sin^K u$, the variable U is generated using $Y \sim \text{Beta}(\frac{K+1}{2}, \frac{1}{2})$. Reversing the power transformation $Z = (U/\pi)^{1/S}$ completes the proof. □

Proof of Theorem 2.2.1.

$$\begin{aligned}
f(x_1 \dots x_n) &= \frac{1}{d^n A^n(s,k)} \left[\prod_{j=1}^{n} \sin\pi\left(\frac{x_j - a}{d}\right)^s\right]^k \\
&= \frac{1}{d^n A^n(s,k)} \left[\prod_{j=1}^{n} \sin\theta_j\right]^k
\end{aligned}$$

Now, the product term can be simplified as:

$$\begin{aligned}
\prod_{j=1}^{n} \sin\theta_j &= \prod_{j=1}^{n}\left(\frac{e^{i\theta_j} - e^{-i\theta_j}}{2}\right) \\
&= \frac{1}{2^n}\sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1}\prod_{j=1}^{n} r_j e^{ir_j\theta_j} && [\because \text{Claim B.1.1}] \\
&= \frac{1}{2^n}\sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1} r\exp\left(i\sum_{j=1}^{n} r_j\theta_j\right) && [\text{let } r := \prod_{j=1}^{n} r_j, \text{ the “product parity”}]
\end{aligned}$$

Plugging the expression back in:

$$f(x_1 \dots x_n) = \frac{1}{A^n(s,k)2^{nk}d^n}\left(\sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1} r\exp\left[i\sum_{j=1}^{n} r_j\theta_j\right]\right)^k$$

The term $\sum_{r_1=\pm1}\cdots\sum_{r_n=\pm1} r\exp\left[i\sum_{j=1}^{n} r_j\theta_j\right]$ is of particular interest. Consider n points $z_j =$

$\frac{x_j - a}{d}$ $[j = 1(1)n]$ in $[0,1]$. Construct the angles $\theta_j = \pi z_j^s$ which lie in $[0,\pi]$. Choose a direction vector $\vec{r} = (r_1 \cdots r_n) \in \{\pm 1\}^n$, where every direction is $r_i = \pm 1$, i.e. either positive or negative. Consider the complex number (corresponding to every choice of a direction vector $\vec{r}$) created by consecutive angular displacements θ_j in the respective directions r_j. Such a complex number is further compensated in sign by the "product parity" r of that direction vector $\vec{r}$. If we sum all such complex numbers created by all possible choices of directions $\vec{r}$, then we get exactly the aforementioned expression. □

Claim B.1.1.

$$\prod_{j=1}^{n} (\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}) = \sum_{r_1=\pm 1} \cdots \sum_{r_n=\pm 1} \prod_{j=1}^{n} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j}$$

Proof. We prove it by induction. For $n = 2$:

$$\begin{aligned}
\prod_{j=1}^{2} (\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}) &= \mathrm{e}^{\mathrm{i}(\theta_1+\theta_2)} - \mathrm{e}^{\mathrm{i}(\theta_1-\theta_2)} - \mathrm{e}^{\mathrm{i}(\theta_2-\theta_1)} + \mathrm{e}^{-\mathrm{i}(\theta_1+\theta_2)} \\
&= \sum_{r_1 \in \{1,-1\}} \sum_{r_2 \in \{1,-1\}} r_1 r_2 \cdot \mathrm{e}^{\mathrm{i}r_1\theta_1} e^{\mathrm{i}r_2\theta_2} \\
&\equiv \sum_{r_1,r_2=\pm 1} \prod_{j=1}^{2} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j}
\end{aligned}$$

Induction hypothesis is $\forall\, n \geq 2$:

$$P(n): \quad \prod_{j=1}^{n} (\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}) = \sum_{r_1=\pm 1} \cdots \sum_{r_n=\pm 1} \prod_{j=1}^{n} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j}$$

We prove it for $n + 1$ case:

$$\begin{aligned}
\prod_{j=1}^{n+1} (\mathrm{e}^{\mathrm{i}\theta_j} - \mathrm{e}^{-\mathrm{i}\theta_j}) &= \left(\sum_{r_1=\pm 1} \cdots \sum_{r_n=\pm 1} \prod_{j=1}^{n} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j} \right) \cdot (\mathrm{e}^{\mathrm{i}\theta_{n+1}} - \mathrm{e}^{-\mathrm{i}\theta_{n+1}}) \\
&= \sum_{r_1=\pm 1} \cdots \sum_{r_n=\pm 1} \prod_{j=1}^{n} r_j \cdot (\mathrm{e}^{\mathrm{i}\theta_{n+1}} - \mathrm{e}^{-\mathrm{i}\theta_{n+1}}) \mathrm{e}^{\mathrm{i}r_j\theta_j} \\
&= \sum_{r_1=\pm 1} \cdots \sum_{r_n=\pm 1} \prod_{j=1}^{n} r_j \cdot \sum_{r_{n+1}=\pm 1} r_{n+1} \mathrm{e}^{\mathrm{i}r_{n+1}\theta_{n+1}} \mathrm{e}^{\mathrm{i}r_j\theta_j} \\
&= \sum_{r_1=\pm 1} \cdots \sum_{r_{n+1}=\pm 1} \prod_{j=1}^{n+1} r_j \mathrm{e}^{\mathrm{i}r_j\theta_j}
\end{aligned}$$

Hence proved. □

Proof of Theorem 2.2.3.

$$\begin{aligned}
\int_0^1 \sin^k(\pi t)\mathrm{d}t &= \int_0^\pi \sin^k\theta\frac{\mathrm{d}\theta}{\pi} \\
&= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \sin^k\theta\mathrm{d}\theta \\
&= \frac{2}{\pi}\cdot 2^{k-1}\mathrm{B}\left(\frac{k+1}{2},\frac{k+1}{2}\right) && \text{[Zwillinger, Eq. 3.621 (1)]} \\
&\equiv \frac{2^k}{\pi}\frac{\Gamma^2\left(\frac{k+1}{2}\right)}{\Gamma(k+1)}
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\int_0^1 \sin^k(\pi t)\mathrm{d}t &= 2\int_0^{\frac{\pi}{2}} \sin^k\theta\frac{\mathrm{d}\theta}{\pi} \\
&\equiv \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \sin^k\theta\cos^0\theta\mathrm{d}\theta \\
&= \frac{2}{\pi}\cdot\frac{1}{2}\mathrm{B}\left(\frac{k+1}{2},\frac{1}{2}\right) && \text{[Zwillinger, Eq. 3.621 (5)]} \\
&= \frac{1}{\pi}\frac{\Gamma\left(\frac{k+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)} \\
&\equiv \frac{1}{\sqrt{\pi}}\frac{\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{k}{2}+1\right)}
\end{aligned}$$

$\square$

Proof of Theorem 2.2.4.

$$\int_0^z \sin^k(\pi t)\mathrm{d}t = \frac{1}{\pi}\int_0^{\pi z}\sin^k\theta\mathrm{d}\theta \quad \left[\theta := \pi t \implies \mathrm{d}t = \frac{\mathrm{d}\theta}{\pi}\right]$$

$$= \frac{1}{\pi}\cdot\left.\begin{cases}\frac{1}{2^{2n}}\binom{2n}{n}\theta + \frac{(-1)^n}{2^{2n-1}}\sum_{k=0}^{n-1}(-1)^k\binom{2n}{k}\frac{\sin[2(n-k)\theta]}{2(n-k)} & [k=2n] \\ \frac{1}{2^{2n}}(-1)^{n+1}\sum_{k=0}^{n}(-1)^k\binom{2n+1}{k}\frac{\cos(2n+1-2k)\theta}{2n+1-2k} & [k=2n+1]\end{cases}\right|_{\theta=0}^{\pi z}$$

[Zwillinger, Eq. 2.513-1, 2]

$$= \frac{1}{\pi}\cdot\begin{cases}\frac{1}{2^{2n}}\binom{2n}{n}\pi z + \frac{(-1)^n}{2^{2n-1}}\sum_{k=0}^{n-1}(-1)^k\binom{2n}{k}\frac{\sin[2\pi(n-k)z]}{2(n-k)} & [k=2n] \\ \frac{1}{2^{2n}}(-1)^{n}\sum_{k=0}^{n}(-1)^k\binom{2n+1}{k}\frac{1-\cos(2n+1-2k)\pi z}{2n+1-2k} & [k=2n+1]\end{cases}$$

$\square$

Proof of Theorem 2.3.1.

$$\begin{aligned}
f(x;d,k) &= \frac{1}{dA(1,k)} \sin^k \left[\pi \frac{x+\frac{d}{2}}{d}\right] \\
&= \frac{\sqrt{\pi}\cdot\Gamma(k+1)}{d\cdot\Gamma\left(\frac{k+1}{2}\right)} \cos^k\left(\frac{\pi x}{d}\right) && \text{[Equation 2.8]} \\
&\sim \frac{\sqrt{\pi}\cdot k^{k+\frac{1}{2}}\mathrm{e}^{-k}}{d\cdot\left(\frac{k-1}{2}\right)^{k/2}\mathrm{e}^{-\frac{k-1}{2}}} \exp\left[k\ln\cos\left(\frac{\pi x}{d}\right)\right] && \text{[Stirling approximation for large } k\text{]} \\
&\sim \frac{\sqrt{\pi}\cdot k^{k+\frac{1}{2}}\left(\frac{\mathrm{e}}{2}\right)^{-\frac{k}{2}}}{d\cdot k^{k/2}} \exp\left[k\ln\left(1-\frac{\pi^2x^2}{2d^2}+\mathcal{O}(d^{-4})\right)\right] && \text{[Maclaurin series in } k \text{ for cos]} \\
&\sim \frac{\sqrt{\pi}}{d} k^{\frac{k+1}{2}} \left(\frac{\mathrm{e}}{2}\right)^{-\frac{k}{2}} \exp\left[-\frac{\pi^2 k x^2}{2d^2} + k\mathcal{O}(d^{-4})\right] && \text{[Mercator series for ln]}
\end{aligned}$$

Let us set $d = \pi\sqrt{k}$, so that:

$$\begin{aligned}
f(x;d,k) &\sim \frac{1}{\sqrt{\pi}} k^k \left(\frac{\mathrm{e}}{2}\right)^{-\frac{k}{2}} \exp\left[-\frac{x^2}{2} + k\mathcal{O}(k^{-2})\right] \\
&\propto \exp\left[-\frac{x^2}{2} + \mathcal{O}(k^{-1})\right] \\
&\to \mathcal{N}(0,1) && \text{as } k\to\infty
\end{aligned}$$

$\square$

B.2 Properties involving the GHGF

Remark B.2.1 (Additional notes on GHGFs). For simplicity, all HGFs (e.g. Confluent HGF) expressed in this paper are represented in the Generalized form only.

Note than if $\mathrm{F}_j :\equiv \frac{(a_1)^{\overline{j}}\cdots(a_p)^{\overline{j}}}{(b_1)^{\overline{j}}\cdots(b_q)^{\overline{j}}}\frac{x^j}{j!}$ denotes the j^{th} term ($j = 0(1)\infty$) of the summation, so that $\mathrm{F}_q^p \equiv \sum_{j=0}^{\infty} \mathrm{F}_j$, then the ratio of consecutive terms of a GHgF defines its parameters and arguments:

$$\frac{\mathrm{F}_{n+1}}{\mathrm{F}_n}\left[\begin{matrix} a_1\cdots a_p \\ b_1\cdots b_1\end{matrix}; z\right] = \frac{(a_1+n)\cdots(a_p+n)}{(b_1+n)\cdots(b_q+n)}\frac{z}{n+1}$$

In our paper, we will also need the Generalized Binomial expansion formula because k may not be integer. Define from hereon $\binom{r}{k} = \frac{(r)^{\overline{k}}}{k!}$ as the generalized binomial coefficient. Then $(x+y)^r = \sum_{k=0}^{\infty}\binom{r}{k}x^{r-k}y^k$, which is the same series as the finite expansion obtained for $r \in \mathbb{Z}^+$, and otherwise an infinite series.

Proof of Theorem 2.4.1.

$$\begin{aligned}
\sin^k(\pi z^s) &= \sin^k \theta && [\text{let } \theta := \pi z^s] \\
&= \left(\frac{\mathrm{e}^{i\theta} - \mathrm{e}^{-\mathrm{i}\theta}}{2i}\right)^k && \because \mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta \\
&= \frac{1}{(2\mathrm{i})^k} \sum_{n=0}^{\infty} \binom{k}{n} \mathrm{e}^{\mathrm{i}\theta(k-n)} (-\mathrm{e})^{-\mathrm{i}\theta n} \\
&= \frac{1}{(2\mathrm{i})^k} \sum_{n=0}^{\infty} (-1)^n \binom{k}{n} \mathrm{e}^{\mathrm{i}\theta(k-2n)} \\
&\equiv \frac{1}{(2\mathrm{i})^k} \sum_{n=0}^{\infty} (-1)^n \binom{k}{n} \mathrm{e}^{\mathrm{i}(k-2n)\pi z^s}
\end{aligned}$$

□

Proof of Theorem 2.4.2. For the first expression, we work with the Maclaurin series of $\sin\theta$ in θ:

$$\begin{aligned}
\sin^k(\pi u) &= (\pi u)^k \left[\sum_{j=0}^{\infty} \frac{(-1)^j (\pi u)^{2j}}{(2j+1)!}\right]^k \\
&= (\pi u)^k \left[\sum_{j=0}^{\infty} \frac{\left(-\frac{\pi^2 u^2}{4}\right)^j}{\left(\frac{3}{2}\right)^{\bar{j}} j!}\right]^k && \because \left(\frac{3}{2}\right)^{\bar{j}} = \frac{(2j+1)!}{2^{2j} j!} \\
&= (\pi u)^k \left(\mathrm{F}_1^0 \left[\begin{matrix} \cdot \\ \frac{3}{2} \end{matrix}; -\frac{\pi^2 u^2}{4}\right]\right)^k
\end{aligned}$$

Alternatively, we construct the required HGF from the ratio of its terms, taking inspiration from Remark B.2.1. Note that if we were to express the infinite summation in Equation 2.11 as a HGF, then the ratio of its terms would give:

$$\frac{\mathrm{F}_{n+1}}{\mathrm{F}_n} = (k+n) \frac{-\exp(-2\mathrm{i}\pi z^s)}{n+1}$$

This should be equivalent to the following HGF:

$$\mathrm{F}_0^1 \left[\begin{matrix} k \\ \cdot \end{matrix}; -\exp(-2\mathrm{i}\pi z^s)\right]$$

We find that a dual representation is possible, but not much reconciliation or interpretation to this duality is found as of yet. □

Proof of Theorem 2.4.3.

$$\begin{aligned}
\varphi_Z(t) &= \int_0^\infty \mathrm{e}^{\mathrm{i}tz}\psi(z;s,k)\mathrm{d}z = \frac{1}{A(s,k)}\int_0^\infty \mathrm{e}^{\mathrm{i}tz}\varsigma(z;s,k)\mathrm{d}z \\
&= \frac{1}{A(s,k)}\int_0^1 \mathrm{e}^{\mathrm{i}tz}\frac{1}{(2i)^k}\sum_{j=0}^{\infty}\binom{k}{j}(-1)^j\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \text{[Equation 2.11]} \\
&= \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\int_0^1 \mathrm{e}^{\mathrm{i}tz}\mathrm{e}^{\pi(k-2j)z^s}\mathrm{d}z && \text{(B.4)}
\end{aligned}$$

We evaluate the inner integral $\mathcal{I}_1 := \int_0^1 \mathrm{e}^{\mathrm{i}tz}\mathrm{e}^{\pi(k-2j)z^s}\mathrm{d}z$ as follows:

$$\begin{aligned}
\mathcal{I}_1 &= \int_0^1\left(\sum_{m=0}^{\infty}\frac{(\mathrm{i}t)^m z^m}{m!}\right)\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \text{[Maclaurin series for } \mathrm{e}^{\mathrm{i}tz} \text{ in } z\text{]} \\
&= \sum_{m=0}^{\infty}\frac{(\mathrm{i}t)^m}{m!}\int_0^1 z^m\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z && \left[\text{Exchanging } \sum \text{ and } \int\right] \quad \text{(B.5)}
\end{aligned}$$

We again evaluate the inner integral $\mathcal{I}_2 := \int_0^1 z^m\mathrm{e}^{\mathrm{i}\pi(k-2j)z^s}\mathrm{d}z$, by writing $\omega := \pi(k-2j)$ for convenience and substituting $u = z^s$, thereby $\implies z = u^{1/s}$ and $\mathrm{d}z = \frac{1}{s}u^{1/s-1}\mathrm{d}u$:

$$\begin{aligned}
\mathcal{I}_2 &= \int_0^1 (u^{1/s})^m\mathrm{e}^{\mathrm{i}\omega u}\frac{1}{s}u^{1/s-1}\mathrm{d}u \\
&= \frac{1}{s}\int_0^1 u^{\frac{m+1}{s}-1}\mathrm{e}^{i\omega u}\mathrm{d}u \\
&= \frac{1}{s}\frac{1}{\frac{m+1}{s}}\,\mathrm{F}_1^1\left[\begin{matrix}\frac{m+1}{s} \\ \frac{m+1}{s}+1\end{matrix};\mathrm{i}\omega\right] && \text{NIST DLMF, 13.2.E4, 13.4.E1} \quad \text{(B.6)}
\end{aligned}$$

Hence, substituting the formulations Equation B.5, Equation B.6 back into the parent equation Equation B.4, we have –

$$\varphi_Z(t) = \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{m=0}^{\infty}\frac{(\mathrm{i}t)^m}{m!(m+1)}\mathrm{F}_1^1\left[\begin{matrix}\frac{m+1}{s} \\ \frac{m+1}{s}+1\end{matrix};\mathrm{i}\pi(k-2j)\right]$$

which completes the proof. □

Proof of Theorem 2.4.4. Since it is known that $E[Z^r] = \frac{1}{\mathrm{i}^r}\left[\frac{\mathrm{d}^r}{\mathrm{d}t^r}\varphi_Z(t)\right]_{t=0}$, by r^{th} order differentiation of the Characteristic Function in Equation 2.14, we obtain:

$$\upmu_r(s,k) = \frac{1}{\mathrm{i}^r}\cdot\frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}\frac{(-1)^j\binom{k}{j}(\mathrm{i}^r r!)}{(r+1)!}\mathrm{F}_1^1\left[\begin{matrix}\frac{r+1}{s} \\ \frac{r+1}{s}+1\end{matrix};\mathrm{i}\pi(k-2j)\right]$$

$$\left[\because \left.\frac{\mathrm{d}^r}{\mathrm{d}t^r}(\mathrm{i}t)^m\right|_{t=0} = \begin{cases}\mathrm{i}^r r! & \text{if } m = r \\ 0 & \text{o.w.}\end{cases}\right]$$

This proves the first expression. Alternatively, by direct integration of the PDF:

$$
\begin{aligned}
\int z^r \sin^k(\pi z^s)\mathrm{d}z &= \int z^r \frac{1}{(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\mathrm{e}^{\mathrm{i}(k-2j)\pi z^s}\mathrm{d}z && [\because \text{Equation 2.11}]\\
&= \frac{1}{(2\mathrm{i})^k}\int \sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi z^s]^n}{n!}z^r\mathrm{d}z && [\text{Maclaurin series for } \mathrm{e}^{\theta} \text{ in } \theta]\\
&= \frac{1}{(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi]^n}{n!}\int z^{ns+r}\mathrm{d}z && \left[\text{Substitute } \int \text{ and } \sum\right]
\end{aligned}
\tag{B.7}
$$

$$
\therefore \mu_r(s,k) = \frac{1}{A(s,k)}\int_0^1 z^r \sin^k(\pi z^s)\mathrm{d}z = \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+r+1)}
$$

Note that the second summation is precisely the term F_1^1. □

Proof of Theorem 2.4.5. This follows from the calculation of moments in Theorem 2.4.4:

$$
\begin{aligned}
\Psi(z;s,k) &= \frac{1}{A(s,k)}\int_0^z \sin^k(\pi t^s)\mathrm{d}t\\
&= \frac{1}{A(s,k)}\frac{1}{(2\mathrm{i})^k}\sum_{j=0}^{\infty}\binom{k}{j}(-1)^j\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi]^n}{n!}\int_0^z t^{ns}\mathrm{d}t \quad [\because \text{Equation B.7, putting } r=0]\\
&= \frac{1}{A(s,k)(2\mathrm{i})^k}\sum_{j=0}^{\infty}(-1)^j\binom{k}{j}\sum_{n=0}^{\infty}\frac{[\mathrm{i}(k-2j)\pi]^n}{n!(ns+1)}z^{ns+1}
\end{aligned}
$$

□

Appendix C

Applications

C.1 Neural Network

Output: $a^{n+1}_{i_{n+1}} \equiv \hat{y}_n$ for $n \in 1(1)N{=}10$
Single-layer network. Middle layer: $z_m = \sum_p w_{pn} x_p + b_n$
Activation: $\sigma(\cdot)$, can be anything.
Loss: Average Cross-entropy (deviance) per batch for a single epoch. $R(\theta) = -\sum_{n=1}^{N} \sum_{p=1}^{P} y_{pn} \ln f_n(x_p)$.

`F.cross_entropy` combines LogSoftmax and NLLLoss (Negative Log Likelihood Loss) in one step. It is the standard choice for multi-class classification problems.
Averaging: Since `F.cross_entropy` (by default) reduces each batch to a single mean value, and we are dividing by `len(train_loader)`, the result is the average loss across all batches processed.

C.1.1 GELU and SiLU activation functions

The Gaussian Error Linear Unit (GELU) activation function is defined as

$$\mathrm{GELU}(x) = x\Phi(x),$$

where $\Phi(x)$ denotes the cumulative distribution function of the standard normal distribution:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2}\, dt.$$

An often-used approximation is

$$\mathrm{GELU}(x) \approx 0.5x \left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}\left(x + 0.044715x^3\right)\right)\right).$$

The GELU activation smoothly weights inputs according to their magnitude and is widely used in transformer-based architectures.

The Sigmoid Linear Unit (SiLU), also known as the Swish activation, is defined as

$$\mathrm{SiLU}(x) = x\sigma(x),$$

where

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

is the logistic sigmoid function. Explicitly,

$$\mathrm{SiLU}(x) = \frac{x}{1 + e^{-x}}.$$

The SiLU activation combines linear and sigmoid behaviour in a smooth and differentiable manner and has demonstrated strong empirical performance in deep neural networks.

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Baidurya Bhattacharyya
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Indian Statistical Institute, 2026.